

DUAL DIETARY ANTIGEN EXPOSURE AS A CONVERGENT DRIVER OF HYPERTENSION AND LIPID DYSREGULATION:

A Hypothesis Linking Neu5Gc Xenosialitis and Gluten-Mediated Intestinal Permeability to Chronic Vascular Inflammation, Impaired Nitric Oxide Bioavailability, and the Atherogenic Lipid Phenotype

Martin Hamer, RN (Ret.)

Independent Researcher | Hamer Glycan Systems | hamerglycans.com | 2026

Abstract

Background: Hypertension affects approximately 1.28 billion adults worldwide and remains the single largest attributable risk factor for cardiovascular mortality. Despite decades of research, a substantial proportion of cases — particularly those without identifiable secondary causes or classical risk factor profiles — remain mechanistically unexplained. A growing body of evidence implicates chronic low-grade systemic inflammation as an upstream driver of both hypertension and atherogenic dyslipidemia, yet the specific dietary sources of that inflammation have received inadequate attention in cardiovascular research.

Hypothesis: We propose that two widely consumed dietary antigens — N-glycolylneuraminic acid (Neu5Gc) from red meat and bovine dairy, and wheat-derived gliadins and amylase-trypsin inhibitors (ATIs) — act through distinct but convergent mechanisms to sustain chronic vascular inflammation, impair nitric oxide (NO) bioavailability, and produce the atherogenic lipid phenotype characterised by hypertriglyceridemia, reduced HDL-C, and small dense LDL predominance. Neu5Gc xenosialitis drives immune complex-mediated endothelial activation and NF- κ B-dependent suppression of endothelial nitric oxide synthase (eNOS). Gliadin-mediated intestinal permeability enables systemic translocation of bacterial lipopolysaccharide (LPS), activating TLR4-driven innate immune pathways that compound the vascular inflammatory state. These two antigen pathways are not merely additive — they interact pharmacokinetically, with gluten-mediated permeability amplifying the systemic bioavailability of dietary Neu5Gc and producing a combined vascular effect exceeding either pathway alone.

Clinical Significance: The patient phenotype most relevant to this hypothesis — hypertension with the atherogenic lipid triad, elevated hsCRP, and blood pressure difficult to control despite conventional measures, in the absence of severe obesity or classical risk factors — may be presenting to clinicians daily without the dietary antigen component of their condition being recognised or addressed. This paper provides a mechanistic framework for that recognition and a set of testable predictions to guide clinical investigation.

Keywords: *hypertension; dyslipidemia; Neu5Gc; xenosialitis; gliadin; ATIs; intestinal permeability; endothelial dysfunction; nitric oxide; eNOS; NF-κB; LPS; atherogenic lipid phenotype; small dense LDL; cardiovascular inflammation; dietary antigen*

1. Introduction: The Unexplained Residual

Consider the patient who presents with stage 1 or 2 hypertension, a lipid panel showing elevated triglycerides, reduced HDL-C, and an LDL that is numerically borderline but flagged as small and dense on particle sizing. Their hsCRP is mildly elevated. They are not obese, or are only modestly so. Their sodium intake is not excessive. They exercise. They do not smoke. They have a moderate family history but nothing that fully accounts for the severity of their cardiovascular risk profile. Their blood pressure responds partially to antihypertensive medication but not as completely as the prescribing physician would like. They are managed, but not well-managed.

This patient is not rare. They sit in cardiovascular clinics and general practice offices across the Western world in enormous numbers. And in the great majority of cases, the dietary component of their inflammatory state is assessed through the lens of sodium, saturated fat, total calories, and fibre — a framework that is not wrong but is, we propose, incomplete. It does not ask whether two specific dietary antigens present in virtually every meal they eat are sustaining a state of chronic vascular inflammation that conventional cardiovascular risk factor modification cannot fully resolve, because it does not address the mechanism producing the inflammation.

The association between unprocessed red meat consumption and cardiovascular disease risk — independent of saturated fat and sodium content — has been documented across multiple large prospective cohorts and has persisted after adjustment for the full range of conventional confounders. The residual risk associated with red meat, once fat, sodium, processing, and lifestyle factors are accounted for, represents an unexplained signal that the conventional framework has not resolved. Similarly, the association between non-celiac gluten sensitivity and elevated cardiovascular inflammatory markers has been documented in individuals who show no celiac autoimmunity and whose symptoms would not traditionally prompt cardiovascular investigation.

We propose that Neu5Gc — present in red meat and bovine dairy — and wheat-derived ATIs and gliadins are the mechanistic explanation for these residual associations: dietary antigens that sustain vascular inflammation through pathways that are not addressed by standard cardiovascular risk factor management, and that interact with each other in ways that amplify their combined vascular impact.

2. Neu5Gc and the Xenosialitis Mechanism

2.1 The Evolutionary Foundation

N-glycolylneuraminic acid (Neu5Gc) is a sialic acid produced by most mammals via the enzyme CMP-Neu5Ac hydroxylase (CMAH). Humans carry an inactivating deletion in the CMAH gene that arose approximately 2–3 million years ago, rendering Neu5Gc immunologically foreign. Despite the inability to synthesise Neu5Gc endogenously, humans regularly ingest it through consumption of red meat (beef, lamb, pork) and bovine dairy products, where it is present at concentrations of 5–30 mg/kg in meat and 7–11 mg/L in bovine milk.

Upon ingestion, dietary Neu5Gc is absorbed at the intestinal mucosa and metabolically incorporated into human cell-surface glycoproteins — a phenomenon confirmed in human-like Cmah-knockout mouse models and in direct human tissue analysis. Because the human immune system has matured in the complete absence of Neu5Gc, it recognises the incorporated molecule as a xenoantigen and mounts an antibody response. Anti-Neu5Gc IgG antibodies are detectable in the serum of most adult humans at concentrations proportional to habitual red meat and dairy consumption.

2.2 Xenosialitis at the Vascular Endothelium

The vascular endothelium is among the tissues most continuously and directly exposed to circulating immune effectors. Endothelial cells express sialylated glycoproteins on their luminal surface — the surface most directly bathed in circulating blood and its immunological contents. Dietary Neu5Gc incorporates into these endothelial glycoproteins, creating neoepitopes at the vascular surface that circulating anti-Neu5Gc IgG antibodies recognise and bind.

The resulting immune complex formation at the endothelial surface activates the classical complement pathway, generating C3b opsonins and the anaphylatoxins C3a and C5a. Fc receptor-bearing leukocytes — monocytes, neutrophils, and NK cells — are recruited to the endothelial surface by these complement activation products. Local release of TNF- α , IL-6, IL-1 β , and IL-8 follows. This constitutes endothelial activation: upregulation of ICAM-1, VCAM-1, and E-selectin; increased leukocyte adhesion; heightened coagulation tendency; and sustained pro-inflammatory gene expression driven by NF- κ B activation in the endothelial cell itself.

Varki and colleagues have termed this process xenosialitis, and in an elegant series of experiments using CMAH-knockout mice — which, like humans, lack endogenous Neu5Gc but incorporate dietary Neu5Gc — Samraj et al. demonstrated that feeding a Neu5Gc-rich diet to mice harbouring anti-Neu5Gc antibodies significantly accelerated atherosclerosis compared to control diets. The mechanism was not a direct lipid effect; it was chronic systemic and vascular inflammation mediated by the immune response to incorporated Neu5Gc. This is the animal model closest to the human scenario, and its findings are directly relevant to the vascular hypothesis presented here.

2.3 The Nitric Oxide Impairment Cascade

The connection between endothelial xenosialitis and elevated blood pressure is substantially mediated through impaired nitric oxide (NO) bioavailability. Endothelial nitric oxide synthase (eNOS) is the primary source of vascular NO, which acts as the principal endothelium-derived vasodilator — relaxing vascular smooth muscle, inhibiting platelet aggregation, suppressing leukocyte adhesion, and maintaining the anti-thrombotic character of the endothelial surface.

Chronic inflammatory cytokine exposure in xenosialitis — particularly TNF- α and IL-1 β — suppresses eNOS expression and activity through NF- κ B-mediated transcriptional repression. Simultaneously, NF- κ B activation upregulates NADPH oxidase subunits, increasing the generation of superoxide at the endothelial surface. Superoxide reacts with NO at a diffusion-limited rate to form peroxynitrite — a reactive nitrogen species that is neither a vasodilator nor a productive signalling molecule, but that instead nitrosylates and inactivates eNOS, uncouples the enzyme to produce more superoxide rather than NO, and oxidises tetrahydrobiopterin (BH4), the essential eNOS cofactor whose depletion further impairs NO synthesis.

The result is a self-reinforcing cycle: xenosialitis-driven NF- κ B activation reduces NO production and increases superoxide generation; superoxide destroys the NO that remains and further uncouples eNOS; the resulting NO deficit reduces vasodilatory tone and raises vascular resistance; and elevated vascular resistance increases shear stress on the endothelium, promoting further NF- κ B activation. Elevated blood pressure is the haemodynamic consequence. In an individual consuming bovine Neu5Gc daily at every meal, this cycle is maintained continuously — not as an acute insult that resolves, but as a persistent background state against which conventional antihypertensive medication must work at a disadvantage.

3. Gluten, Intestinal Permeability, and Systemic LPS Translocation

3.1 The Gliadin-Zonulin Pathway

Wheat gluten consists primarily of glutenins and gliadins — storage proteins that resist complete proteolytic digestion in the human gastrointestinal tract. Gliadin-derived peptides interact with intestinal epithelial cells through the chemokine receptor CXCR3, triggering the release of zonulin, the primary physiological regulator of intestinal tight junction assembly. Zonulin-mediated disruption of ZO-1, occludin, and claudin-1 — the structural proteins of the tight junction complex — increases paracellular permeability, allowing luminal contents that are normally excluded to pass between epithelial cells into the subepithelial space and subsequently the systemic circulation.

This is not a mechanism limited to patients with celiac disease. Fasano and colleagues have demonstrated gliadin-induced zonulin release and measurable increases in intestinal permeability in both celiac and non-celiac subjects, distinguishing the two groups by the magnitude and persistence of the permeability increase rather than its presence or absence. Non-celiac gluten sensitivity (NCGS) patients demonstrate elevated serum markers of intestinal barrier disruption — including fatty acid-binding protein 2 (FABP2), a biomarker of enterocyte damage — consistent with genuine gliadin-mediated barrier compromise in the absence of celiac autoimmunity.

3.2 Metabolic Endotoxaemia: LPS as a Vascular Inflammatory Driver

The most immunologically consequential consequence of increased intestinal permeability for vascular health is the enhanced systemic translocation of bacterial lipopolysaccharide (LPS)

from the colonic lumen. LPS is the outer membrane component of gram-negative bacteria present in enormous quantities in the normal colonic microbiome. Under conditions of normal intestinal barrier integrity, LPS is largely excluded from systemic circulation. Under conditions of gliadin-induced permeability, even sub-nanogram circulating concentrations of LPS activate TLR4/MD-2 signalling on monocytes, macrophages, endothelial cells, and adipocytes, triggering NF- κ B-mediated transcription of TNF- α , IL-6, and IL-1 β .

Cani and colleagues introduced the term 'metabolic endotoxaemia' to describe this state of chronic, sub-clinical LPS-driven innate immune activation. In animal models, chronic low-dose LPS infusion produces a metabolic and inflammatory phenotype — including weight gain, insulin resistance, elevated inflammatory markers, and endothelial dysfunction — that mirrors the changes observed in individuals consuming high-fat, high-gluten Western diets. The mechanistic link between intestinal permeability, LPS translocation, and cardiovascular inflammatory risk is increasingly well-supported and represents a parallel inflammatory input to the xenosialitis mechanism that is activated by a different dietary antigen through a different receptor system but converges on the same vascular downstream effectors.

3.3 ATIs as Independent TLR4 Activators

Amylase-trypsin inhibitors (ATIs) — non-gluten wheat proteins distinct from the gliadin fraction — are potent direct activators of TLR4 on innate immune cells. Junker and colleagues demonstrated that ATIs drive intestinal and mesenteric lymph node myeloid cell activation at nanogram concentrations, producing TNF- α , IL-6, and IL-12 output that is independent of LPS and independent of celiac autoimmunity. ATI-mediated TLR4 activation does not require intestinal permeability to produce systemic inflammatory effects — it operates in the gut-associated immune compartment and distributes its cytokine output systemically through the lymphatic and portal circulation.

The combined inflammatory input of ATI-driven TLR4 activation and gliadin-mediated LPS translocation represents a two-pronged wheat-derived vascular inflammatory burden that operates through distinct but converging mechanisms and is present in any individual consuming wheat-containing products on a regular basis.

4. Convergence and Interaction: More Than the Sum of Parts

4.1 Shared Downstream Effectors

Neu5Gc xenosialitis and wheat-derived metabolic endotoxaemia converge on a shared set of downstream vascular effectors: NF- κ B activation in endothelial cells, upregulation of ICAM-1, VCAM-1, and E-selectin, suppression of eNOS, amplification of NADPH oxidase-driven superoxide generation, and the consequent impairment of NO bioavailability and elevation of vascular resistance. Neither pathway alone is sufficient to account for the full cardiovascular risk profile of many patients. Together, activating both simultaneously, they produce a combined endothelial activation state that is qualitatively more sustained and more resistant to resolution than either pathway alone.

The two pathways activate overlapping but distinct inflammatory gene expression programmes — xenosialitis primarily through adaptive immune mechanisms (IgG-mediated complement activation and Fc receptor engagement), metabolic endotoxaemia primarily through innate pattern recognition receptor signalling (TLR4). Their simultaneous engagement may produce inflammatory output that exceeds what either antigen alone would generate through shared downstream mediators — IL-6-induced acute-phase protein synthesis, TNF- α -mediated sensitisation of TLR signalling, and the compounding effects of concurrent complement activation and TLR4-driven cytokine release on endothelial NF- κ B tone.

4.2 The Permeability-Neu5Gc Pharmacokinetic Interaction

The most novel mechanistic interaction proposed in this paper concerns the pharmacokinetics of Neu5Gc absorption itself. Under conditions of normal intestinal barrier integrity, dietary Neu5Gc entering the gut lumen is absorbed primarily through transcellular routes — passage through enterocytes — which are subject to cellular processing that may partially attenuate the intact glycan structures before they reach the systemic circulation.

Under conditions of gliadin-mediated paracellular permeability — where the tight junctions between enterocytes are functionally compromised — Neu5Gc-bearing glycoproteins and free Neu5Gc from the gut lumen gain access to the paracellular route: direct passage between enterocytes into the subepithelial space without cellular processing. This paracellular route deposits Neu5Gc-bearing material directly into the lamina propria in a form more structurally intact — and therefore potentially more immunogenically active — than the transcellularly processed form.

The practical consequence is a pharmacokinetic amplification loop: an individual consuming both red meat or dairy (Neu5Gc source) and wheat (permeability driver) simultaneously may be absorbing a higher effective Neu5Gc antigenic load than would be predicted from the dietary Neu5Gc content alone. The permeability that wheat creates becomes the enhanced absorption route for the Neu5Gc that meat and dairy deliver. The two antigens are not merely additive in their inflammatory effects — one is creating the conditions for the other to be more efficiently absorbed and more immunologically potent.

This pharmacokinetic interaction has not been directly measured. It is mechanistically plausible, consistent with known intestinal permeability physiology, and represents a tractable experimental question — paracellular versus transcellular Neu5Gc absorption under conditions of gliadin-induced tight junction disruption is directly measurable in intestinal epithelial models. If confirmed, it would mean that the Western dietary pattern — in which red meat and wheat are routinely consumed together at the same meal — produces a Neu5Gc antigenic burden at the systemic level that is substantially greater than the Neu5Gc content of the meat alone would predict.

4.3 The Clinical Picture: What This Looks Like in Practice

The patient whose vascular risk is substantially driven by the dual-antigen mechanism described in this paper has a recognisable clinical profile, even if the mechanism itself is not currently part of the diagnostic framework:

Feature	Conventional Attribution	Dual-Antigen Interpretation
Hypertension resistant to conventional management	Inadequate medication dose; non-adherence; secondary causes	Persistent NO impairment from ongoing xenosialitis and LPS-driven eNOS suppression; cannot fully resolve while dietary antigen load continues
Atherogenic lipid triad (high TG, low HDL, sdLDL predominance)	High carbohydrate or fat intake; metabolic syndrome; genetic dyslipidaemia	Inflammation-driven VLDL overproduction (TNF- α); apoA-I suppression (IL-6); CETP-mediated LDL remodelling to sdLDL
Mildly elevated hsCRP without obvious inflammatory disease	Subclinical infection; obesity; unexplained	Chronic xenosialitis and/or ATI-driven TLR4 activation maintaining low-grade NF- κ B tone
Cardiovascular risk disproportionate to classical risk factor burden	Familial predisposition; unmeasured risk factors	Dietary antigen-driven vascular inflammation operating through pathways not captured by conventional risk calculators
GI sensitivity without celiac diagnosis	IBS; FODMAP sensitivity; functional GI disorder	ATI-mediated TLR4 activation and gliadin-induced permeability producing the gut symptoms as a parallel manifestation of the same antigenic burden driving the vascular risk

Table 1. Clinical Profile of the Dual-Antigen Vascular Phenotype. These features, taken together in the same patient, warrant consideration of dietary xenoantigen load as a contributing mechanism — and dietary elimination as a low-risk diagnostic and therapeutic trial.

5. Inflammatory Remodelling of Lipid Metabolism

5.1 The Atherogenic Lipid Phenotype as an Inflammatory Signature

The atherogenic lipid phenotype — elevated fasting and postprandial triglycerides, reduced HDL-C, and predominance of small dense LDL particles — is increasingly understood as an inflammatory metabolic signature rather than primarily a dietary fat phenomenon. It is this inflammatory mechanism, rather than dietary saturated fat alone, that connects the dual-antigen hypothesis to lipid dysregulation.

The inflammatory pathway to atherogenic dyslipidaemia operates through three well-characterised steps. First, TNF- α — chronically elevated in both xenosialitis and ATI/LPS-driven innate activation — stimulates hepatic VLDL production while simultaneously impairing adipose tissue lipoprotein lipase (LPL) activity, reducing peripheral VLDL clearance and producing sustained hypertriglyceridaemia. Second, IL-6 suppresses hepatic synthesis of apolipoprotein

A-I (apoA-I), the primary structural and functional protein of HDL particles, reducing HDL-C production and impairing reverse cholesterol transport. Third, elevated circulating triglycerides drive cholesteryl ester transfer protein (CETP)-mediated neutral lipid exchange between VLDL and LDL, depleting LDL of cholesteryl esters and enriching it with triglycerides; subsequent hepatic lipase activity hydrolyses these triglyceride-enriched LDL particles into the smaller, denser, more electronegative LDL subspecies that are more susceptible to oxidation, more readily retained in the arterial wall, and more potently atherogenic.

5.2 Why This Lipid Pattern Does Not Fully Respond to Standard Interventions

The atherogenic lipid phenotype driven by chronic dietary antigen inflammation has a feature that distinguishes it clinically from the classical saturated fat-driven hypercholesterolaemia for which standard lipid management was designed: it is inflammatory in origin, not substrate-driven. Reducing saturated fat intake addresses the substrate available for LDL synthesis but does not attenuate the TNF- α -driven VLDL overproduction, the IL-6-driven apoA-I suppression, or the CETP-mediated sdLDL remodelling that are the mechanistic drivers of the triad. Statins reduce LDL-C effectively but have modest effects on triglycerides and limited effects on the sdLDL particle profile that is most predictive of cardiovascular events in this phenotype. Fibrates address triglycerides but do not resolve the underlying inflammatory driver.

A patient with inflammation-driven atherogenic dyslipidaemia who eliminates the dietary antigen sources of that inflammation may experience lipid improvements that no amount of statin or fibrate therapy could produce — because those medications are not addressing the mechanism. This is not a rejection of pharmaceutical management; statins and fibrates remain valuable tools. It is a recognition that pharmaceutical tools designed for substrate-driven dyslipidaemia cannot substitute for removal of an inflammatory driver that those tools were not designed to address.

6. Testable Predictions

Prediction	Investigation Design	Expected Finding
Simultaneous high Neu5Gc and high ATI/gliadin dietary burden will produce higher vascular inflammatory markers than either antigen alone	Crossover dietary intervention: four arms (high Neu5Gc only; high ATI/gliadin only; both; neither) matched for total calories and macronutrients; primary outcomes: hsCRP, IL-6, TNF- α , soluble VCAM-1, anti-Neu5Gc IgG; secondary: ambulatory blood pressure, fasting lipid panel with LDL particle sizing	Highest inflammatory marker elevation in combined-exposure arm; compounding effect exceeding additive prediction consistent with pharmacokinetic interaction hypothesis
Dual dietary elimination will produce measurable	Double-blind placebo-controlled dietary elimination trial in stage	Reduced systolic and diastolic BP; reduced triglycerides; improved HDL-

Prediction	Investigation Design	Expected Finding
reductions in ambulatory blood pressure and atherogenic lipid markers over 8–12 weeks	1–2 hypertension with atherogenic lipid phenotype; primary: ambulatory BP at 8 and 12 weeks; secondary: triglycerides, HDL-C, sdLDL particle number by NMR, hsCRP, anti-Neu5Gc titres	C; shift from sdLDL to large-buoyant LDL predominance; reduced hsCRP and anti-Neu5Gc titres
Anti-Neu5Gc antibody titres and intestinal permeability markers will correlate positively with BP variability and sdLDL particle number after adjustment for conventional CVD risk factors	Cross-sectional study in hypertensive and normotensive adults; serum anti-Neu5Gc IgG; urinary lactulose:mannitol ratio; serum zonulin; FABP2; 24-hour ambulatory BP; LDL particle number by NMR; multivariate regression adjusting for age, BMI, sodium intake, physical activity	Independent positive associations between anti-Neu5Gc titres and BP variability; between permeability markers and sdLDL particle number; associations persist after adjustment for conventional risk factors
Pharmacological restoration of intestinal barrier function will attenuate xenosialitis markers in the context of a Neu5Gc-rich diet	CMAH-knockout mouse model: Neu5Gc-rich diet with and without butyrate supplementation or tight-junction stabilising agent; primary: serum anti-Neu5Gc IgG, tissue Neu5Gc incorporation, vascular TNF- α , aortic atherosclerotic lesion burden	Reduced Neu5Gc incorporation and xenosialitis severity with barrier restoration; supporting the permeability-Neu5Gc pharmacokinetic interaction hypothesis
Individuals with genetic variants associated with enhanced anti-Neu5Gc antibody production or heightened zonulin response to gliadin will show disproportionately elevated BP and atherogenic lipid burden on Western dietary patterns	Biobank linkage study: pharmacogenomic variants in Fc receptor genes (FCGR2A, FCGR3A), complement pathway genes, and zonulin pathway regulators linked to dietary pattern data and cardiometabolic phenotype data	Enrichment of high-risk variants in hypertensive and atherogenic dyslipidaemia cohorts; interaction between variant and dietary Neu5Gc/gluten burden on cardiovascular phenotype

Table 2. Testable Predictions of the Dual Dietary Antigen Vascular Hypothesis. Prediction 1 directly tests the pharmacokinetic interaction — the most novel claim of this paper — through a four-arm crossover design that can isolate the compounding effect.

7. Limitations and Alternative Interpretations

The direct human evidence for Neu5Gc xenosialitis as a driver of vascular disease remains largely inferential at the clinical level, with the mechanistic work performed predominantly in carefully constructed but ultimately murine models. The magnitude of Neu5Gc incorporation into

human vascular tissue under habitual dietary conditions has not been precisely quantified in a population-representative sample, and it remains possible that the quantities incorporated in typical dietary consumers are insufficient to generate the immune complex density required for meaningful complement activation at the endothelial surface.

The gliadin-permeability hypothesis faces the additional challenge that the causal direction of the zonulin-permeability relationship remains contested. Some investigators argue that increased intestinal permeability is a consequence rather than a cause of systemic inflammatory states, and that elevated zonulin in inflammatory conditions reflects downstream mucosal immune activation rather than upstream barrier disruption. The diagnostic reliability of commercially available serum zonulin assays has also been questioned, with concerns about cross-reactivity and assay standardisation that have not yet been fully resolved.

Red meat and wheat consumption are highly correlated with other dietary and lifestyle variables that independently influence cardiovascular risk, and rigorous isolation of the hypothesised antigen-specific effects from confounding by total caloric intake, dietary fibre, omega-3 fatty acid content, physical activity, and sleep quality presents a formidable methodological challenge in human observational studies. The crossover dietary design proposed in Prediction 1 is the most rigorous available approach to this isolation but requires careful macronutrient matching to prevent the elimination of red meat and wheat from producing secondary nutritional changes that confound interpretation.

This hypothesis does not claim that Neu5Gc and ATIs/gliadins are the primary causes of hypertension or atherogenic dyslipidaemia at the population level. The classical risk factors — sodium, saturated fat, obesity, physical inactivity, genetic predisposition, and hormonal factors — remain the dominant drivers in most cases. The claim is more specific: that for a definable subset of susceptible individuals, particularly those with the clinical profile described in Section 4.3, these dietary antigens constitute a significant and currently unrecognised contributing mechanism whose removal may produce cardiovascular improvements that conventional management cannot achieve alone.

8. A Note on the Acnerase Platform

The Acnerase food-label scanning application, documented in the Hamer Loop Manuscript Series, was originally conceived as a dietary screening tool for the cutaneous inflammatory conditions — acne and rosacea — that served as the originating clinical observations of the Neu5Gc/ATI framework. Its ingredient identification logic, however, is antigen-specific rather than condition-specific: it identifies Neu5Gc sources, ammonium catalysts, and ATI/gliadin-containing ingredients regardless of the downstream condition for which that information is sought.

For individuals with the vascular risk profile described in this paper — hypertension with atherogenic dyslipidaemia, elevated hsCRP, and dietary Neu5Gc and gluten burden as a suspected contributing mechanism — Acnerase provides the same practical ingredient identification function that it provides for acne patients: real-time identification of hidden dietary

antigen sources in commercial food products, across the full range of processed food label synonyms that make avoidance difficult without specialised knowledge. The tool does not require the user to have an inflammatory skin condition. It requires only that they wish to identify and reduce their dietary Neu5Gc and AIT/gliadin burden — for whatever reason that goal is relevant to them.

9. Conclusion

Hypertension and atherogenic dyslipidaemia are treated, in the great majority of cases, with pharmaceutical tools designed for the downstream consequences of vascular inflammation — antihypertensives that manage elevated vascular resistance, statins that reduce LDL-C, fibrates that address triglycerides. These tools are effective and important. They are also, for a subset of patients, insufficient — because they are not addressing the upstream inflammatory driver that is producing the vascular resistance and the lipid dysregulation in the first place.

The vascular endothelium is not a passive conduit. It is an active immunological interface continuously sampling the antigenic composition of the circulating blood. When that blood carries daily boluses of anti-Neu5Gc immune complexes from bovine dietary Neu5Gc, and daily LPS translocation from wheat-induced intestinal permeability, the endothelium responds with the activation state — NF- κ B-driven, NO-depleting, atherogenic — that this paper describes. It is doing what an endothelium does when faced with chronic antigenic challenge. The challenge just happens to arrive in a bun with a burger, every day, at every meal, in ways that current dietary cardiovascular risk assessment does not capture.

The hypothesis is testable. The predictions are specific. The intervention — dietary elimination of the two antigen sources — is low-risk, zero-cost, and accessible to any motivated patient without a prescription. For the patient sitting in the cardiovascular clinic whose blood pressure will not fully respond to medication and whose lipid panel stubbornly maintains the atherogenic pattern despite all that conventional management can offer — this question deserves to be asked. What are you eating? And is what you are eating sustaining the very inflammation that your medication is struggling to overcome?

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